

Assignment #3**Name: Abdullah Gamal Moustafa Elafifi****ID: 20221238****Summarized Report:**

Firstly, I initialized a population of chromosomes as a randomly generated set of real-valued pairs within a defined range $[-2, 2]$. The fitness evaluation was based on the given function:

$$\text{Maximize } F(x_1, x_2) = 8 - (x_1 + 0.0317)^2 + (x_2)^2, \text{ where } -2 \leq x_1, x_2 \leq 2$$

Selection was performed using **tournament selection**, where individuals competed in groups of size k , and the fittest were chosen. **Arithmetic crossover** was applied with a defined probability to generate offspring, allowing smooth blending of parent genes. Finally, **Gaussian mutation** introduced small variations to maintain diversity. **Elitism** ensured that the top-performing individuals were carried over to the next generation unchanged.

Experimental Desing (as requested in the assignment):

- Population Size: **20**
- Chromosome Representation: **Real-valued pairs (X1, X2)**
- Tournament Selection (varying k): **Two values of k tested**
- Chromosome Length: **5**
- Generations: **100**
- Crossover Probability: **0.6**
- Mutation Probability: **0.05**
- Elitism: **Comparison between with and without elitism**

The Best Fitness Values:

The best fitness per generation was recorded to analyze the improvement in solution quality

over time. Both **elitism** and **non-elitism** cases were compared to evaluate their impact with:
K = 8 and K=6

Getting these results →

Results for k=3 (with elitism):

Best Fitness: 11.78998693978573

Average Fitness: 11.834720740936142

Results for k=30 (with elitism):

Best Fitness: 12.0

Average Fitness: 11.87620548071895

Results without elitism:

Best Fitness: 12.0

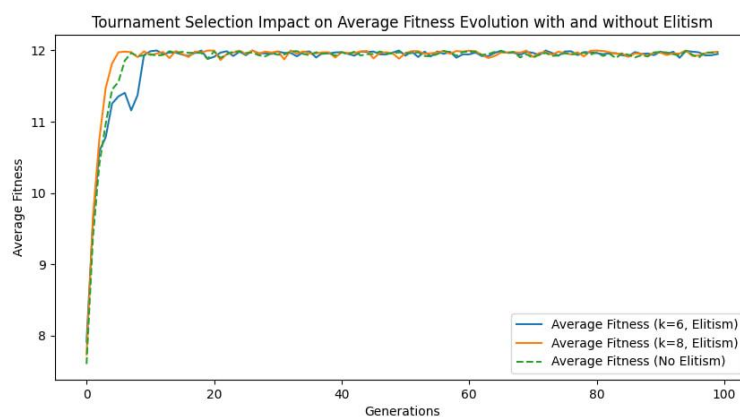
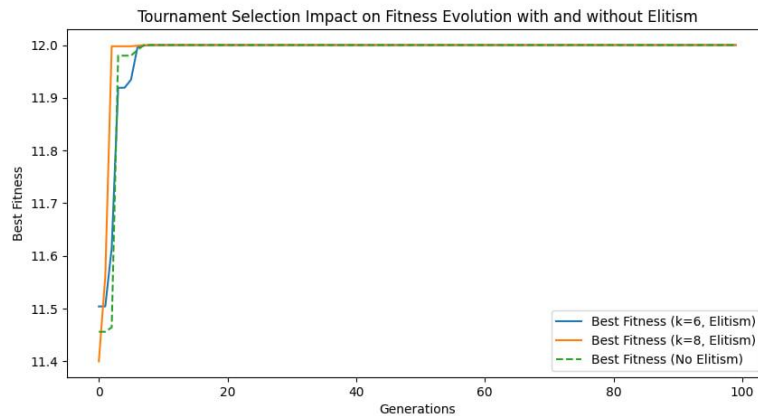
Average Fitness: 11.846986555873793

Tools and Technologies:

- Python
- Numpy Library
- Random Library
- Matplotlib (for visualizations).

I have put a random seed to ensure the reproducibility of the code per each run:

`np.random.seed( X )`



Do you have any improvements over Assignment #2's results?

I think this assignment shows improvements over Assignment #2 due to the implementation of **real encoding, arithmetic crossover, Gaussian mutation, and tournament selection**. Unlike the previous assignment, which used binary and Gray encoding, real encoding provides a more precise search space exploration, leading to faster convergence and better fitness values.

Additionally, tournament selection (especially with a higher k) **demonstrated stronger selection pressure**, helping to retain fitter individuals in the population. The inclusion of elitism further ensured that the best solutions were preserved across generations, leading to higher final fitness scores.

The comparison of elitism vs. non-elitism confirmed that elitism improves both the best and average fitness values over generations. The plots illustrate that elitism maintains steady improvements, whereas non-elitism results in greater fluctuations and potential loss of good solutions.

Which of the above operators is the most effective one (Selection, Crossover, or Mutation) in improving the results?

Selection (Tournament Selection with varying k):

- A higher k value (stronger selection pressure) led to better convergence as it consistently -favored stronger individuals.
- A lower k value (weaker selection pressure) introduced more diversity but resulted in slower convergence.
- *The results showed that increasing k helped find better solutions faster while maintaining diversity.*

Crossover (Arithmetic Crossover):

Helped in exploiting good solutions by combining parent characteristics effectively. However, it was dependent on selection, meaning if poor individuals were selected, crossover could not significantly improve results.

Mutation (Gaussian Mutation):

Added randomness and prevented premature convergence by introducing slight variations. While useful, its impact was secondary compared to selection and crossover.

Conclusion and Analysis:

The population gradually evolved towards higher fitness values. In early generations, there was a wider diversity in solutions. Over time, selection pressure, crossover, and mutation guided the population toward more optimal solutions.

Tournament selection (higher k) ensured that fitter individuals were consistently chosen, leading to faster convergence.

Elitism preserved the best solutions, preventing the loss of good individuals and resulting in higher final fitness values.

Non-elitism led to more fluctuations, demonstrating the importance of retaining top solutions across generations.

The results confirmed that a combination of strong selection pressure and elitism significantly improved performance, producing a higher-quality solution compared to a purely random evolutionary process.

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